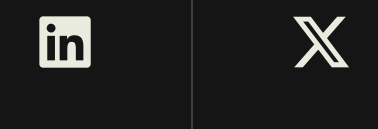
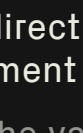
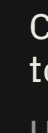
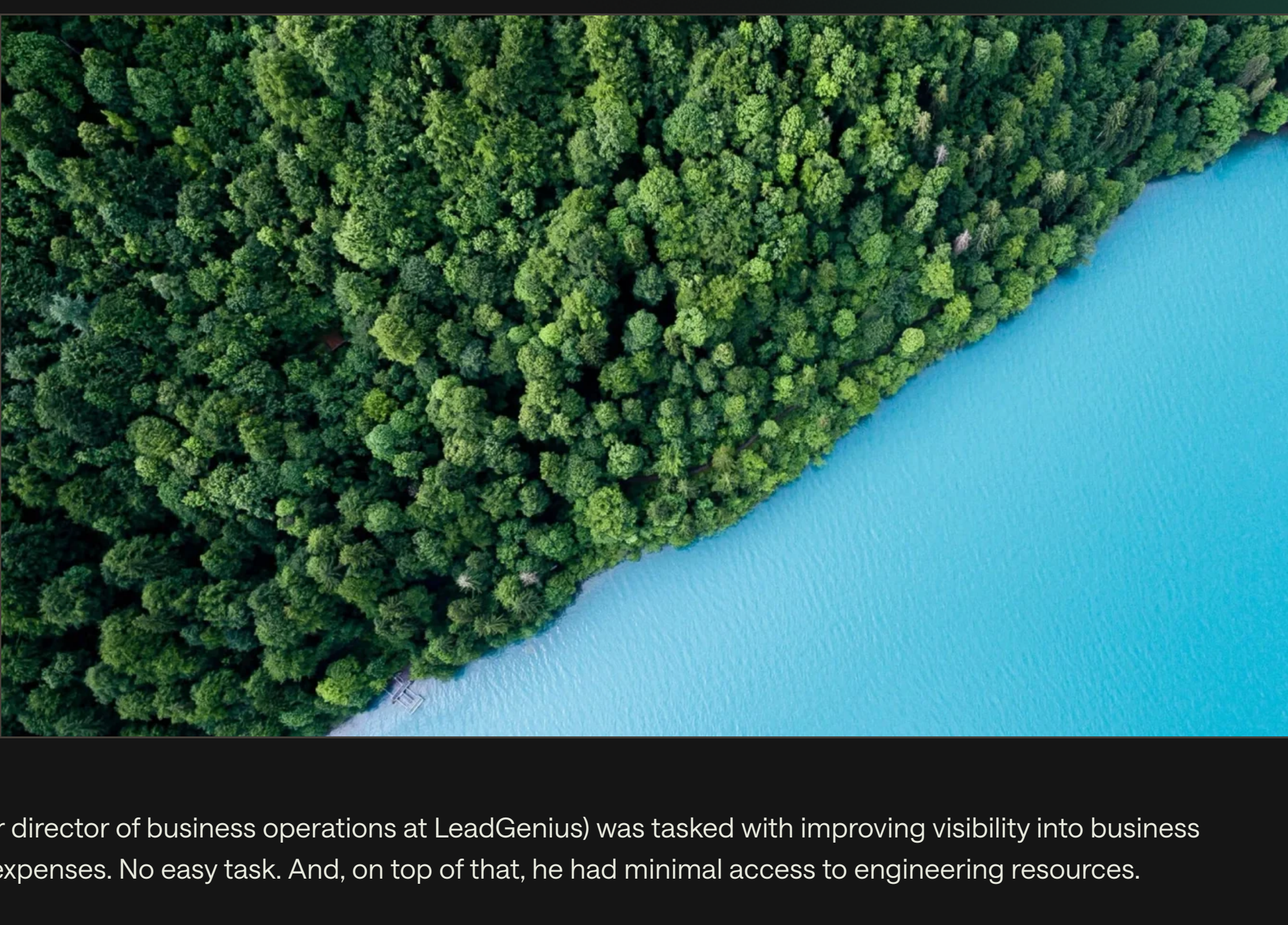


The cost-benefit analysis of internal tools



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- Calculating direct investment in internal tool development
- Uncovering the value of internal tools
- Uncovering the actualized value of internal tooling

Adam Louie (senior director of business operations at LeadGenius) was tasked with improving visibility into business costs and cutting expenses. No easy task. And, on top of that, he had minimal access to engineering resources.

To build out a robust suite of internal tools to give the LeadGenius team insight into their business, Adam turned to Retool. The Retool apps Adam built unlocked visibility into their finances and helped them identify cost-cutting patterns that saved them around \$1M in just one year.

Internal tools like the ones Adam built provide essential visibility into data and productivity for teams. However, they also require commitment and investment from the teams who are building *and* using them.

The decision to invest in internal tools is all about striking the right balance. In most cases, that balance lies between the engineering overhead to develop (and maintain) internal tools and the workforce productivity those tools deliver once they're up and running.

Drawing upon our recent survey of *The State of Internal Tools in 2020*, we thought it would help shine a light on what to consider when investing in building internal tools.

Calculating direct investment in internal tool development

Let's start with the most direct cost of internal tools: **the cost of development**. Simply put, we can calculate this by taking an engineer's pay per hour multiplied by the number of hours it takes to build and maintain the tool. Or $\$/\text{hour} \times \text{project-hours}$.

According to Glassdoor ⁷, the national average pay for a software engineer is roughly \$92,000 a year. Now, if you're in an industry that uses many custom internal tools (say, financial services and logistics), the math for this is pretty simple. You'd probably have at least one (maybe more) software engineers working full time on internal tools, so you'd just multiply that average by however many full-time folks you'd need.

To calculate hard costs (what you pay engineers for developing internal tools), identify an engineer's hourly pay, and multiply that by the hours per week spent on internal tool development or maintenance. Using our Glassdoor estimate from before, a conservative \$92,000 a year salary is roughly \$44 per hour. Maybe one of your engineers spends three months (480 hours) building up an internal tool, then an additional 10 hours per week updating and maintaining it for the rest of the year. That would mean 600 hours $\times$ \$44 = \$26,400.

If you're just getting started, looking at benchmark data can provide a useful starting point for estimating engineering personnel required for internal tool development. According to a survey about the state of internal tools we conducted earlier this year, teams at companies with more than 20 employees spend more than 20% of their time building and maintaining internal tools. And as headcount grows, that number jumps to close to 40% for teams at companies with more than 1,000 employees.

Percentage of team's time spent on internal tools by company size

As companies get larger, internal tooling takes up more of their time



The cost of development is the most direct cost of internal tools and is the easiest to quantify. *But what about the indirect costs of not investing in internal tools?* That's where we'll turn to next.

Uncovering the value of internal tools

The flip-side of determining the investment in internal tools is looking at the value they bring. When instrumented the right way, internal tools add a ton of value, like uniting multiple back-office systems, reducing the probability of errors, and strengthening security. All of these things bring improvements to the workflows of people who use the tools, which helps them focus on higher-value work, like keeping customers happy.

As a hypothetical, let's say your company is an online retailer that sells dog identification tags. You have a team of customer support agents who have access to three different programs:

1. An **order management system (OMS)** to track orders from customers
2. A spreadsheet to personalize dog identification tags and export as a CSV for the manufacturer
3. Customer support software where agents can communicate with customers via phone, text, or email

One day, a customer emails customer support because the dog identification tag they ordered showed up reading "Fido" instead of "Dido." To amend the situation, your customer support agent must first go from the support software to the order management system to look at the customer's order. They have to pivot to the dog-tag design spreadsheet to customize a new tag, add that order to the OMS, and then email the customer back with their updated information. All of this context switching and delay in processing orders introduces more potential for error. It also results in an overall worse experience for your workforce and customers.

It's hard to put a precise number on the cost of not using internal tools (what's the cost of an unhappy customer or annoyed support agent?). Instead of calculating these intangibles, we thought it would be helpful to share a few examples of the value organizations see as a result of investing in internal tools.

Neo4j reduced churn by moving renewals out of Salesforce

Mike Brophy (VP of Global Renewals Sales at Neo4j) **built a renewals control center (RCC)** to accurately forecast and hit renewal targets.

Before building RCC, Mike would ask his reps for account updates during their regular meetings. He had no way to see renewal timelines in one place or track down statuses without involving his team, which felt like he was chasing information or interrupting his reps' days just to get a pulse on their accounts.

With the RCC, Mike can easily track the stage of each renewal and even filter by rep. He now has a clear understanding of the work his team is doing and knows where to deploy reps to make sure renewals are completed without a hitch. What's more, the RCC gives every rep a broad overview of their accounts, complete with extra details that were often left out of Salesforce profiles. These might include the reason for churn, contract details, or personal notes about a contact.

Noble Schools built an app to keep 12,500 students engaged during a pandemic

Moon Lee (Chief Data and Innovation Officer at Noble Schools) needed a application to take remote attendance, conduct wellness checks, track barriers to distance learning for the 2020-2021 school year.

Moon built a **Multi-Tiered System of Support (MTSS)**, a tool used within Noble's education systems to track the learning and well-being of students. MTSS is used to track attendance and as a report generating tool that tracks barriers to learning and overall performance to assist teachers in identifying students who need extra help.

LeadGenius saved \$1M with better internal tooling

As mentioned in the intro, Adam Louie (Senior Director of Business Operations at LeadGenius) built three apps with Retool to give his teams real-time visibility into the data they previously couldn't.

User Profile Editor: An app to view, clean, and organize customer accounts

Community Costs: An app that tracks contractor cost metrics and displays real-time reports on revenue

Project Manager Dashboard: An app that enables Project Managers to organize their data campaign workflows within a homegrown CRM.

We built Retool to help developers build internal tools remarkably fast. Retool removes the tedious parts of development like wrestling with UI libraries, stitching together data sources, and configuring access controls. In turn, engineering teams can ship internal tools 10x faster and focus more time building the core product. [Try Retool for free here.](#)

While it may be challenging to determine (in dollars and cents) the actual cost of not having internal tools, the right custom tool can certainly unlock a ton of value. For those who have made a committed investment in building custom apps for their workforce, 80% say who say that internal tools are critical to their company's operational success.